D.P.M.S.
Artificial Intelligence

1η Σειρά Ασκήσεων 2025-2026

Course: Natural Language Processing - NLP

Ioannis Petrousov

March, 2026

A. Tokens, Types, Zipf's Law

1

	NLTK	spaCy	Bert
#tokens	93356	97152	117968
#types	11017	10445	8544
TTR	0.1180	0.1075	0.0724
Hapax legomena	5615	5072	2915
Hapax dislegomena	1702	1659	1381

The method which creates the largest vocabulary is the one with the highest #types count, in this case NLTK. However, it's important to note that while this number is the highest of all other methods, it's not necessary the most accurate. Specifically, word_tokenize() uses the Punkt sentence tokenization model to splits the text into individual elements. This model uses a word-based tokenization approach which also splits punctuation marks, including ",", "." and """ into separate tokens thus increasing the #types count. The same can be observed for the tokenization method spaCy uses. This results into creating tokens of the form:

```
In [1]: "." in tokens
Out[1]: True

In [2]: "," in tokens
Out[2]: True

In [3]: "'" in tokens
Out[3]: True

In [4]: "it's" in tokens
Out[4]: False

In [5]: "'s" in tokens
Out[5]: True
```

On the other hand, Bert uses a more modern subword tokenization approach - WordPiece. The method effectively recognizes and constructs larger words (from smaller) thus optimizing #types count.

TTR expresses vocabulary diversity. The higher values observed in NLTK and spaCy indicate a richer vocabulary. However, BERT has a vocabulary consisting of longer words which might have an increased meaning compared to the ones from the aforementioned methods.

Both spaCy and NLTK follow Zipf's law: $5615/11017 = 50\%$ and $5072/10445 = 48\%$ so half of the vocabulary appears only once in the tokens. However, BERT does not follow this law $2915/8544 = 34\%$.

2

I selected the first sentence: ”Pierre Vinken, 61 years old, will join the board as a nonexecutive director Nov. 29.”. Following are the results from the tokenization.

```
NLTK tokens:
['Pierre', 'Vinken', ',', '61', 'years', 'old', ',', 'will',
'join', 'the', 'board', 'as', 'a', 'nonexecutive', 'director',
'Nov.', '29', '.']
===============
spaCy tokens:
['Pierre', 'Vinken', ',', '61', 'years', 'old', ',', 'will',
'join', 'the', 'board', 'as', 'a', 'nonexecutive', 'director',
'Nov.', '29', '.']
===============
BERT tokens:
['Pierre', 'Vin', '##ken', ',', '61', 'years', 'old', ',', 'will',
'join', 'the', 'board', 'as', 'a', 'none', '##xe', '##cut', '##ive',
'director', 'Nov', '.', '29', '.']
===============
```

We can see that both NLTK and spaCy managed to split almost all the words correctly because they use a word-based tokenization approach. However, BERT uses a statistical method to construct tokens. In the example we can see that it broke the word ”Vinken” into 2: ”Vin”, ”##ken”. This happening because the word does not exist in its vocabulary. Also ”nonexecutive” was broken into 4 subwords: ”none”, ”##xe”, ”##cut”, and ”##ive” because it greedy-matched separate words from it’s vocabulary starting with the word ”none”. It couldn’t match the rest of the word (”xecutive”) so it broke it into 3 pieces using ”##” in front of each piece.

3

	NLTK Type	NLTK Freq	spaCy Type	spaCy Freq	BERT Type	BERT Freq
0	,	4823	.	5020	.	6363
1	the	4757	,	4823	,	5026
2	.	3645	the	4763	the	4767
3	of	2318	of	2319	’	4117
4	to	2175	to	2180	of	2320
5	a	1967	a	1991	to	2226
6	in	1760	in	1773	a	2150
7	and	1534	and	1541	in	1940
8	”	959	”	1372	-	1733
9	’s	865	-	1231	and	1549
10	for	851	’s	865	s	1356
11	that	848	for	851	for	870
12	$	708	NaN	<NA>	that	848
13	is	672	NaN	<NA>	NaN	<NA>

I extracted the number of common tokens programmatically and it was found to be 9. See Python notebook for details.

4

5

References

- NLTK word_tokenize src: https://github.com/nltk/nltk/blob/develop/nltk/tokenize/__init__.py#L127
- SpaCy tokenizer docs: https://spacy.io/usage/linguistic-features#how-tokenizer-works
- FreqDist from NLTK: https://www.nltk.org/api/nltk.probability.FreqDist.html
- Pandas DF different dict size: https://www.statology.org/pandas-dataframe-from-dict-with-different-le
- Pandas DF alight text center: https://stackoverflow.com/questions/17232013/how-to-set-the-pandas-data
- Pandas fix column type: https://pandas.pydata.org/docs/user_guide/integer_na.html
- Set intersection https://www.w3schools.com/python/ref_set_intersection.asp